What Makes Audio Latents Mixing-Equivariant? A Controlled Study of Explicit Supervision

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***Abstract*—Mixing is linear on waveforms, so a latent space in which interpolating two codes and decoding reproduces the waveform mix is an *equivariant* representation. Recent work induces this in a consistency autoencoder by decoding averaged latents against the true mix. We ask which part of that supervision is responsible, whether the target architecture limits what it delivers, and whether the certifying metrics measure it at all, through a controlled study in which each experiment intervenes on one factor. An explicit decode-mixing loss suffices: on a waveform GAN autoencoder it makes the latent space mixing-equivariant at no measured reconstruction cost, at $2\times$ tighter compression too, and it restores level as well as shape. An adversarial term on the mixed path adds nothing beyond inflating a decode-consistency metric, exposing that metric as a confounded proxy; on a consistency-model autoencoder the loss improves latent geometry but not waveform arithmetic. Latent stem subtraction, in contrast, is governed by the latent origin, not the loss: adding the encoded silence $f(0)$ brings every model to within 0.5 dB of its reconstruction ceiling on all 49 test tracks. Comparisons are single-run, matched at one initialization.**

***Index Terms*—representation learning, equivariant representations, audio autoencoders, latent arithmetic, controlled ablation, source separation**

I. INTRODUCTION

Audio mixing is a linear operation: $\mathrm{mix}(x_1, x_2, \alpha) = \alpha x_1 + (1-\alpha)x_2$. For an autoencoder with encoder f and decoder g, we call the latent space *mixing-equivariant* when

$$g(\alpha f(x_1) + (1-\alpha) f(x_2)) \approx \alpha x_1 + (1-\alpha)x_2. \qquad (1)$$

Eq. 1 constrains combinations whose coefficients sum to one. Latent *subtraction*, $g(f(x_{\mathrm{mix}})-f(x_{\mathrm{stem}}))$, uses coefficients $(1, -1)$ and so is not implied by it: an offset $f(x) = x + b$, $g(z) = z - b$ satisfies Eq. 1 exactly yet subtracts to $x_{\mathrm{res}} - b$. Subtraction additionally needs the latent map to have no offset, which we treat empirically (Section IV-C). Where it holds, latent arithmetic becomes audio editing without a dedicated separation or synthesis model, and it is a prerequisite for latent-diffusion systems [1], [2] that compose independently modeled sources. Yet standard autoencoders do not satisfy Eq. 1: their objectives reward reconstruction fidelity, not algebraic structure, and quantized codecs (DAC [3], EnCodec [4]) cannot represent interpolated codes at all.

Torres et al. [5] induce linearity in a consistency autoencoder by decoding the *average* of separately encoded latents and training it to reconstruct the true mix, a supervision principle related to mixup [6]. This leaves open which part of that supervision is responsible for the property, whether the encoder or the decoder must carry it, whether a consistency decoder can turn latent linearity into usable waveform arithmetic, and whether the metrics used to certify it measure it at all. We answer these questions with a controlled study: the same principle written as an explicit loss (Section III) on a waveform GAN autoencoder, whose effect we then isolate by intervening on one factor at a time. Our contributions are:

1) A controlled study of mixed-latent supervision on a waveform GAN autoencoder: the decode-mixing loss alone is sufficient, an adversarial term on the mixed path adds no ground-truth gain, and an encoder-only penalty or a frozen encoder identify the decoder as the locus (Section IV-A).
2) Generalization against matched controls: from-scratch training at $2\times$ tighter compression, and a re-run on a consistency-model autoencoder where latent geometry improves but waveform arithmetic does not (Section IV-B).
3) A downstream analysis showing latent stem removal is limited by the latent origin rather than the loss (Section IV-C).
4) A methodological correction: the widely used decode-vs-decode SI-SDR$_{\mathrm{lin}}$ is confounded by decoder phase variance; we report a ground-truth-referenced variant.

II. RELATED WORK

Neural audio codecs. DAC [3], EnCodec [4], and SoundStream [7] pair convolutional encoders with HiFi-GAN decoders [8] and multi-scale discriminators, optimizing reconstruction without latent structure; their residual vector quantization also precludes latent interpolation. RAVE [9] learns a variational latent space but does not target mixing equivariance, and disentanglement regularizers [10] encourage generic smoothness rather than a specific algebraic identity. **Equivariant representations.** Mixing-equivariance follows the symmetry-based view of representation learning [11], though convex mixing is an affine combination, not a group action, so we borrow the framing and not the formalism. Verma and Berger [12] build equivariance into the architecture; we treat it as a property to be *induced* by supervision and isolate which supervision induces it. **Linear audio representations.** Music2Latent [13] is a consistency autoencoder with partial linearity as a byproduct. Torres et al. [5] make it approximately linear by conditioning the decoder on the average of separately encoded latents while denoising the true mix under the unchanged consistency loss (plus a random gain for homogeneity, which we do not study). Our $\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}}$ is the explicit-loss form of that additivity supervision: the same input (averaged latents) and target (true mix), differing in loss family (multi-resolution STFT and mel versus consistency), mixing coefficient ($\alpha \sim U[0,1]$ versus 0.5), and decoder class (deterministic GAN versus consistency model). We add the analysis: which ingredient carries the effect, a matched re-run on their architecture with a continued-training control (Section IV-B), and ground-truth-referenced metrics. Their additivity (SNR, spectral distance) and separation scores compare two decodes, the protocol Section III-B shows to be confounded, and their separation is negative on every stem (-0.3 to -1.8 dB), as our phase-coherence finding predicts.

III. Method

A. Decode-mixing loss

Given a batch $\{x_i\}$ we form pairs (x_i, x_j) with a fixed-point-free permutation and draw $\alpha \sim U[0,1]$. Let $\bar{x} = \alpha x_i + (1-\alpha)x_j$ be the waveform mix and $\bar{z} = \alpha f(x_i)+(1-\alpha)f(x_j)$ the latent interpolation. The decode-mixing loss requires the decoded interpolation to match the true mix:

$$\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}} = \mathcal{L}_{\mathrm{recon}}(g(\bar{z}),\, \bar{x})\,, \tag{2}$$

where $\mathcal{L}_{\mathrm{recon}}$ is the multi-resolution STFT + mel loss also used for reconstruction. Gradients flow through g and back into f via $\bar{z}$, so a single term constrains both networks; it costs one extra decoder pass per step. We also study an *encoder-only* variant that penalizes encoder nonlinearity directly, $\mathcal{L}_{\mathrm{mix}}^{\mathrm{enc}} = \|f(\bar{x}) - \bar{z}\|^2$ (one extra encoder pass, no decoder pass). The total generator objective is $\mathcal{L} = \mathcal{L}_{\mathrm{recon}} + \mathcal{L}_{\mathrm{GAN}} + \lambda\,\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}} + \lambda_{\ell_2}\|z\|^2$.

B. Metrics

We report reconstruction $\text{SI-SDR}_{\mathrm{rec}} = \mathrm{SISDR}(g(f(x)), x)$ and two equivariance views at $\alpha{=}0.5$:

- **SI-SDR**$_{\mathrm{lin}}$ (decode-vs-decode): $\mathrm{SISDR}(g(\bar{z}), g(f(\bar{x})))$, equivariance relative to the model's *own* reconstruction of the mix.
- **SI-SDR**$_{\mathrm{lin}}^{\mathrm{gt}}$ (vs. ground truth): $\mathrm{SISDR}(g(\bar{z}), \bar{x})$, the quantity in Eq. 1 up to scale, comparable across models. SI-SDR fits a gain before scoring, so it does not assess level preservation; we report the fitted gain separately (Section IV-B).

We also report $\ell_{\mathrm{lat}} = \|\bar{z} - f(\bar{x})\|^2/\|f(\bar{x})\|^2$ (encoder linearity) and $\mathrm{MixRate} = \mathcal{L}_{\mathrm{recon}}(g(\bar{z}), \bar{x})/\mathcal{L}_{\mathrm{recon}}(g(f(\bar{x})), \bar{x})$ (decoder equivariance; < 1 is better than the encode-then-decode path). SI-SDR$_{\mathrm{lin}}$ shares the decode-vs-decode structure of prior work's decoder-additivity metrics (SNR and spectral distance between two decodes [5]), and it is *gameable*: we hypothesize that a decoder driven onto a common realistic manifold raises the mutual SI-SDR of its two decode paths without bringing either closer to $\bar{x}$ (Section IV-A). It is

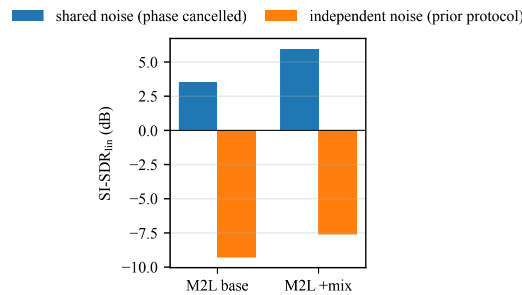

Fig. 1. Decode-noise protocol confound (Music2Latent, MUSDB18, $\alpha{=}0.5$). The same checkpoints score deeply negative under independent-noise decoding and positive under shared-noise decoding; the ≈ -8 dB figure reflects decoder noise, consistent with phase variance, not latent linearity.

also confounded for stochastic decoders: a consistency-model decoder [13] draws fresh noise per call. Decoding the *same* latent twice is deterministic under shared noise but scores -1.8 dB under independent noise, with no latent operation at all, so the decoder's sampling alone accounts for most of the collapse. Fig. 1 shows the same Music2Latent checkpoints scoring ≈ -8 dB under independent-noise decoding (the protocol implied by prior reports) and *positive* under shared-noise decoding, a 12 dB swing driven by the eval protocol rather than the model. This shows protocol sensitivity; decoding the same latent twice with independent noise isolates how much of the disagreement arises without any latent arithmetic (Section IV-B). We therefore treat SI-SDR$_{\mathrm{lin}}^{\mathrm{gt}}$ and MixRate as the equivariance metrics of record, and use shared decode noise everywhere.

C. Experimental setup

The primary model is a waveform autoencoder (DAC-style 1-D convolutional encoder, HiFi-GAN decoder [8], combined multi-scale-STFT and multi-period discriminator) operating at 44.1 kHz on 1-second mono chunks, trained on a ∼950-hour union of FMA-large, MAESTRO, and MUSDB18-HQ [14].[1] Two compression regimes are used: $7.66\times$ ($d{=}128$) and $15.3\times$ ($d{=}64$). Mixing losses are added either as a 25k-step fine-tune over a converged baseline (v2.x) or from scratch over 250k steps (v3.x). All runs use a single NVIDIA RTX 5090. Unless noted, metrics are computed over the full multi-source test set; the α-sweep (Section IV-D) uses a source-balanced subsample.

IV. Results

A. Which signal causes equivariance? An interventional ablation

Table I ablates the loss on the $7.66\times$ GAN autoencoder (25k-step fine-tunes from a shared converged baseline). Each row differs from its comparator in exactly one factor: $\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}}$ and $\mathcal{L}_{\mathrm{mix}}^{\mathrm{enc}}$ against the baseline (the loss), $\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}}$+disc against $\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}}$ (the adversarial term), and frozen-encoder $\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}}$ against

[1]Code, training configurations, pretrained weights, every metric reported here, and audio examples: https://github.com/nurdauletakhanov/musicgen

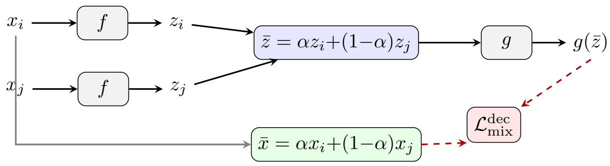

Fig. 2. Decode-mixing loss: encode a pair, interpolate latents, decode, and compare to the true waveform mix. Gradients (dashed) flow through the decoder and back into the encoder via $\bar{z}$.

Loss	$\mathrm{SDR}_{\mathrm{rec}}$	$\mathrm{SDR}_{\mathrm{lin}}$	$\mathrm{SDR}^{\mathrm{gt}}_{\mathrm{lin}}$	ℓ_{lat}	MixRate	FAD
baseline (no mix)	+11.3	+12.0	+8.0	0.068	1.203	0.045
$\mathcal{L}_{\mathrm{dec}}$	+11.4	+12.1	+10.2	0.052	0.972	0.044
$\mathcal{L}_{\mathrm{dec}}$ + disc-on-mix	+11.4	+16.0	+9.9	0.050	1.052	0.044
$\mathcal{L}_{\mathrm{enc}}$ (γ=5)	+11.3	+12.8	+8.4	0.044	1.175	0.044
$\mathcal{L}_{\mathrm{enc}}$ (γ=10)	+11.3	+13.1	+8.5	0.039	1.165	0.044
$\mathcal{L}_{\mathrm{enc}}$ (γ=20)	+11.3	+13.5	+8.7	0.034	1.153	0.045
$\mathcal{L}_{\mathrm{dec}}$ frozen-enc	+11.2	+11.7	+9.5	0.072	1.056	0.046

TABLE I

Mechanism ablation on the GAN autoencoder (7.66×, full test set, α=0.5). $\mathrm{SDR}_{\mathrm{lin}}$ is decode-vs-decode; $\mathrm{SDR}^{\mathrm{gt}}_{\mathrm{lin}}$ is vs. ground truth. Single seed.

Model	$\mathrm{SDR}_{\mathrm{rec}}$	$\mathrm{SDR}_{\mathrm{lin}}$	$\mathrm{SDR}^{\mathrm{gt}}_{\mathrm{lin}}$	MixRate	FAD
GAN AE, 7.66×, no mix	+11.3	+12.0	+8.0	1.203	0.045
GAN AE, 7.66×, +mix	+11.4	+12.1	+10.2	0.972	0.044
GAN AE, 15.3×, no mix	+10.0	+10.6	+6.8	1.187	0.052
GAN AE, 15.3×, +mix	+9.9	+14.6	+8.6	1.046	0.061
M2L, no mix (baseline)	-3.3	+5.0	-9.6	1.101	0.127
M2L, fine-tune control	-3.1	+5.1	-9.5	1.125	–
M2L, +mix	-2.6	+6.9	-7.8	1.086	0.119

TABLE II

Generalization across compression and architecture (full test set, α=0.5, EMA weights for Music2Latent). The M2L "fine-tune control" is the continued-training baseline with mixing disabled.

$\mathrm{SDR}^{\mathrm{gt}}_{\mathrm{lin}}$	FMA	MAESTRO	MUSDB	all
7.66×, no mix	+6.8	+13.8	+6.2	+8.0
7.66×, +mix	+8.9	+16.9	+8.2	+10.2
Δ	**+2.1**	**+3.1**	**+2.0**	**+2.3**
15.3×, no mix	+5.8	+12.0	+5.5	+6.8
15.3×, +mix	+7.2	+15.1	+6.7	+8.6
Δ	**+1.4**	**+3.2**	**+1.2**	**+1.7**

TABLE III

Ground-truth equivariance per test domain (dB). Each Δ is against the matched no-mixing control at the same compression rate. The gain is present on every domain, not carried by one.

$\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$ (the encoder's freedom to adapt); the design is not a full factorial. Three observations drive the paper.

(1) Reconstruction is unaffected. $\mathrm{SDR}_{\mathrm{rec}}$ is flat across all configurations (11.2–11.4 dB), and so is FAD (0.044–0.046): the mixing loss buys equivariance for free, on both a sample-wise and a distributional measure of quality.

(2) The decode-mixing loss is the active ingredient; the discriminator on the mixed path is not. On the ground-truth metric, $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$ alone improves $\mathrm{SDR}^{\mathrm{gt}}_{\mathrm{lin}}$ from 8.0 to 10.2 dB and is the only run with MixRate <1 (0.97). Adding an adversarial term to the mixed path ("+disc-on-mix") does *not* improve ground-truth equivariance (9.9 dB, MixRate 1.05), yet it dramatically inflates the decode-vs-decode $\mathrm{SDR}_{\mathrm{lin}}$ (12.1 → 16.0 dB). The discriminator pulls the two decode paths onto a common realistic manifold (our hypothesis), raising their mutual SI-SDR without moving either toward $\bar{x}$. This is direct evidence that $\mathrm{SDR}_{\mathrm{lin}}$ is gameable, and that given $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$ no adversarial term is needed; whether the discriminator has an effect on its own, without $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$, is not tested here.

(3) The effect is carried through the decoder. The encoder-only variant $\mathcal{L}^{\mathrm{enc}}_{\mathrm{mix}}$ drives ℓ_{lat} lowest but leaves MixRate high (1.15–1.18): it linearizes the encoder without making the decoder equivariant. Conversely, training only the decoder under $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$ with the encoder frozen retains most of the gain (9.5 dB versus 10.2 joint and 8.0 baseline) while leaving encoder linearity unimproved (ℓ_{lat} 0.072). Joint training is best among the tested configurations.

B. Intervening on compression and architecture

Table II extends the recipe along two axes. **Compression:** trained *from scratch* at 15.3× (v3.1), the mixing model matches its no-mixing control (v3.0) on reconstruction (9.9 vs. 10.0 dB) while improving ground-truth equivariance by +1.7 dB ($\mathrm{SDR}^{\mathrm{gt}}_{\mathrm{lin}}$ 6.8 → 8.6), MixRate (1.19 → 1.05), and encoder linearity (ℓ_{lat} 0.078 → 0.057). FAD, however, rises from 0.052 to 0.061 (Table II); whether that is perceptible is untested, so "no quality cost" is established at 7.66× (Table I) and only for $\mathrm{SDR}_{\mathrm{rec}}$ at 15.3×. Because v3.0 is a matched control (same architecture, compression, and schedule, mixing disabled), this isolates the loss's contribution and shows the effect is not an artifact of the looser 7.66× rate or of warm-starting. **Level preservation.** SI-SDR discards gain, so we also report the gain it fits: the decoded mix is 1.6 dB too quiet without the loss and 0.9 dB with it, matching the model's own reconstruction level error (0.8 dB); at 15.3×, 1.8 versus 1.0 against 0.9. The loss restores level as well as shape. **Architecture:** we fine-tune the published Music2Latent [13] consistency autoencoder with the decode-mixing objective, the additivity supervision of [5] on its own architecture, using its native consistency loss on the mixed pair in place of the discriminator. A continued-training control, the same fine-tune with mixing disabled, moves equivariance by <0.2 dB and *worsens* FAD, so the gain is the loss, not domain adaptation. Both metric families agree on the improvement. Absolute $\mathrm{SDR}^{\mathrm{gt}}_{\mathrm{lin}}$ nevertheless stays negative: this configuration improves latent *geometry* (MixRate, ℓ_{lat}) without reaching usable waveform arithmetic. The systems differ in encoder, representation, loss family, compression and training data as well as decoder class, so the contrast isolates none of them as the cause; we report the GAN autoencoder as the primary result and treat this as a limit on transfer, not an explanation of it.

The gain is not domain-driven. Our test set spans three

Model	drums	bass	vocals	other	all	gap↓
7.66× no mix	+4.2	+0.4	+5.2	+3.9	+3.4	5.2
$+f(0)$	+8.8	+6.8	+9.6	+8.0	+8.3	0.4
7.66× $+\mathcal{L}_{\mathrm{dec}}$	+5.7	+4.9	+7.2	+5.1	+5.7	3.0
$+f(0)$	+8.5	+6.9	+9.4	+8.3	+8.3	0.5
7.66× $+\mathcal{L}_{\mathrm{dec}}$+disc	+6.2	+5.4	+7.5	+5.0	+6.0	2.6
$+f(0)$	+8.5	+7.0	+9.3	+8.1	+8.2	0.4
15.3× no mix	+1.2	-1.6	+3.0	+2.1	+1.2	6.6
$+f(0)$	+7.3	+5.8	+8.8	+7.2	+7.3	0.5
15.3× +mix	+5.4	+4.7	+6.9	+3.9	+5.2	2.5
$+f(0)$	+7.3	+6.1	+8.5	+7.2	+7.3	0.4
M2L +mix	-9.9	-12.3	-9.9	-12.7	-11.2	8.4

TABLE IV

STEM REMOVAL VIA LATENT SUBTRACTION (SI-SDR, DB; MUSDB18 TEST). HIGHER IS BETTER; "GAP" IS THE LINEARITY TAX AGAINST EACH MODEL'S OWN ENCODE–DECODE CEILING (LOWER IS BETTER). "$+f(0)$" ROWS ADD THE ENCODED SILENCE TO THE SUBTRACTED LATENT (ORIGIN CORRECTION).

acoustically distinct domains: crowd-sourced indie production (FMA), solo piano (MAESTRO), and multi-track studio mixes (MUSDB18). Table III breaks $\mathrm{SDR}^{\mathrm{gt}}_{\mathrm{lin}}$ out by domain against the matched no-mixing control at each compression rate. The improvement appears on all three at both rates (+2.0 to +3.1 dB at 7.66×; +1.2 to +3.2 dB at 15.3×), so it reflects a property of the learned latent space rather than an artifact of one recording style. The absolute ordering (MAESTRO easiest, MUSDB hardest) tracks reconstruction difficulty and is preserved by the mixing loss.

C. Downstream: latent-arithmetic stem removal

We remove a stem by latent subtraction, $\hat{x} = g(f(x_{\mathrm{mix}}) - f(x_{\mathrm{stem}}))$, and measure SI-SDR against the true residual on the MUSDB18 test set. This is an *oracle* setting: the removed stem is available and encoded, so it tests the representation's arithmetic, not blind separation; its use is latent-domain editing when stem codes exist, as in a pipeline that stores compressed stems or a latent generative model that composes sources without decoding each. Subtraction uses coefficients $(1, -1)$, which Eq. 1 does not cover (Section I), so we report raw subtraction and the origin-corrected form $g(f(x_{\mathrm{mix}}) - f(x_{\mathrm{stem}}) + f(0))$, exact for affine f, g. We also report the *linearity tax* gap $= \mathrm{SISDR}(g(f(x_{\mathrm{res}})), x_{\mathrm{res}}) - \mathrm{SISDR}(\hat{x}, x_{\mathrm{res}})$, the shortfall against the model's own encode–decode ceiling.

Raw subtraction favours the mixing loss; origin correction removes the difference. Raw, $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$ roughly doubles subtraction quality at 7.66× (+3.4 → +5.7 dB) and the matched 15.3× pair goes from +1.2 to +5.2 dB. Adding $f(0)$, one encode of silence, lifts *every* model to within 0.5 dB of its ceiling (Table IV): +2.0 to +6.0 dB per model, on all 49 tracks. With the origin corrected the loss no longer helps subtraction: $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$ versus no mixing is −0.03 dB (95% track-cluster CI [−0.15, +0.11]) and the 15.3× pair +0.01 dB ([−0.19, +0.21]). The raw advantage was the no-mixing models' larger latent offset ($\|f(0)\|$ 6.6 versus 5.4 at 7.66×, 18.1 versus 14.8 at 15.3×), which coefficient-sum-zero arithmetic pays for. What the loss buys is convex equivariance (Tables I and III); subtraction needs the origin, a free post-hoc fix.

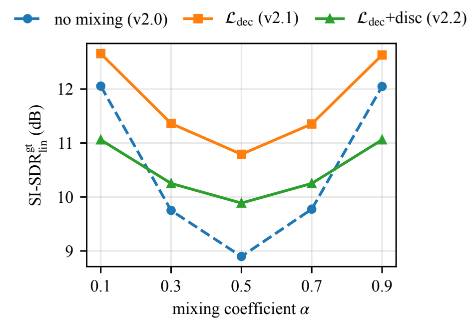

Fig. 3. Ground-truth equivariance vs. mixing coefficient α (source-balanced subsample). The no-mixing baseline sags at the hardest balanced mix (α=0.5); the decode-mixing loss flattens and lifts the curve across all α.

Baselines without latent arithmetic. The unchanged mixture scores +17.1 dB against the residual, above every autoencoder's ceiling, since the removed stem is a small share of the energy: this metric cannot certify removal in absolute terms, only arithmetic relative to the model's own reconstruction. Within the model, autoencoding the mixture and doing nothing scores +2.2 dB (+1.7 at 15.3×); decoding both signals and subtracting waveforms scores +7.5 (+6.5). Raw latent subtraction is below waveform subtraction for every model, origin-corrected latent subtraction above it by +0.7 to +0.9 dB (track-cluster CIs exclude zero), with or without the loss.

All intervals here come from a paired cluster bootstrap over the 49 tracks (10^4 resamples), the correlated unit; they cover evaluation variance at the tested initialization.

D. Equivariance across mixing ratios

Sweeping α on a source-balanced subsample (Fig. 3), the mixing loss *flattens* the equivariance curve: the no-mixing baseline sags most at the hardest balanced mix (α=0.5) and recovers toward the extremes, while mixing models stay near-uniform across α (e.g. $\mathrm{SDR}^{\mathrm{gt}}_{\mathrm{lin}}$ swing of 1.2 dB for $\mathcal{L}^{\mathrm{dec}}_{\mathrm{mix}}$+disc vs. 3.2 dB for the baseline). Thus the property holds across mixing ratios, not only at the trained operating point.

V. CONCLUSION

A controlled study identifies the decode-mixing loss, not an adversarial term on the mixed path, as the signal responsible for convex mixing-equivariance, at no measured reconstruction cost and across compression rates. Raw stem subtraction is instead governed by the latent offset, which one encode of silence removes for every model. On a consistency-model autoencoder the loss improves latent geometry but not waveform arithmetic, so practical claims are confined to the GAN autoencoder we test. **Limitations.** Every comparison is matched at a single initialization, identifying each factor's effect there but not its variation across training runs; experiments are mono; and we conduct no listening test. Stereo, multi-seed estimates, and latent-diffusion composition are future work.

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